

Online supplement to:
*Completion-aware search with partial witness repair for dense
fixed-shape facility layout*

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This supplement contains four secondary experiments referenced from the main text: zero shot transfer (S1), quality and diversity fronts (S2), amortized cost (S3) and action granularity (S4), followed by the full improvement formulation ablation (S5), a qualitative certificate gallery (S6), the complete MCNC/GSRC conversion (S7) and the large instance feasibility record (S8). Each experiment follows the protocol in the main text. The main claims do not depend on these experiments, and we report the relevant control for each experiment. Section S9 gives the full budget sensitivity and mechanism record for the comparison in which all methods start from the same layout. Section S10 collects the diagnostic evidence removed from the shortened main text; complete run-level outputs remain in the archive.

S1 Zero shot transfer

A trained policy can have an advantage through amortization. A solver restarts for every brief, while a policy evaluates a new brief in one forward pass. We trained a policy only on procedurally generated briefs. The type inventory, dimensions, affinity tables, plate aspect ratio, fill ratio, wall preferences and objective constants were randomized during 1.2M training steps. We then evaluated the policy on briefs that were not optimized during training.

Table S1: Zero shot transfer at $n = 32$. The greedy constructor provides the direct comparison because, like the policy, it does not search over the plate. It also requires no training. Its cost is 32 and 528 marginal objective sweeps per layout; the batched figure on the heterogeneous generated set is higher, so we report the cost for one instance.

test set	shelf	SA 45k lat.	greedy constructive		RL zero shot ($\times 16 + \text{pol.}$)
			area ord.	room+pos.	
generated (in distribution)	1 912.6	2 613.9	2 488.8	2 597.5	2 591.0
OFFICE (unseen)	2 942.5	3 412.1	3 359.5	3 380.2	3 426.3
CLINIC (unseen)	3 503.7	4 011.9	3 955.6	4 116.5	3 950.2
evaluations/layout	1	58 963 to 65 875	32	528	2 049 to 5 633

On OFFICE the zero shot policy reaches 3 414.5 with a single greedy decode (32 evaluations, 12.7 ms), matching annealing’s 3 412.1 at 60 371 evaluations, and 3 426.3 with 16 samples and a polish. At the same 32 evaluations the constructor without training reaches 3 359.5.

CLINIC gives a different result. The greedy constructor reaches 4 116.5, which is higher than both full budget annealing and the zero shot policy. At a matched 32 evaluations, it reaches

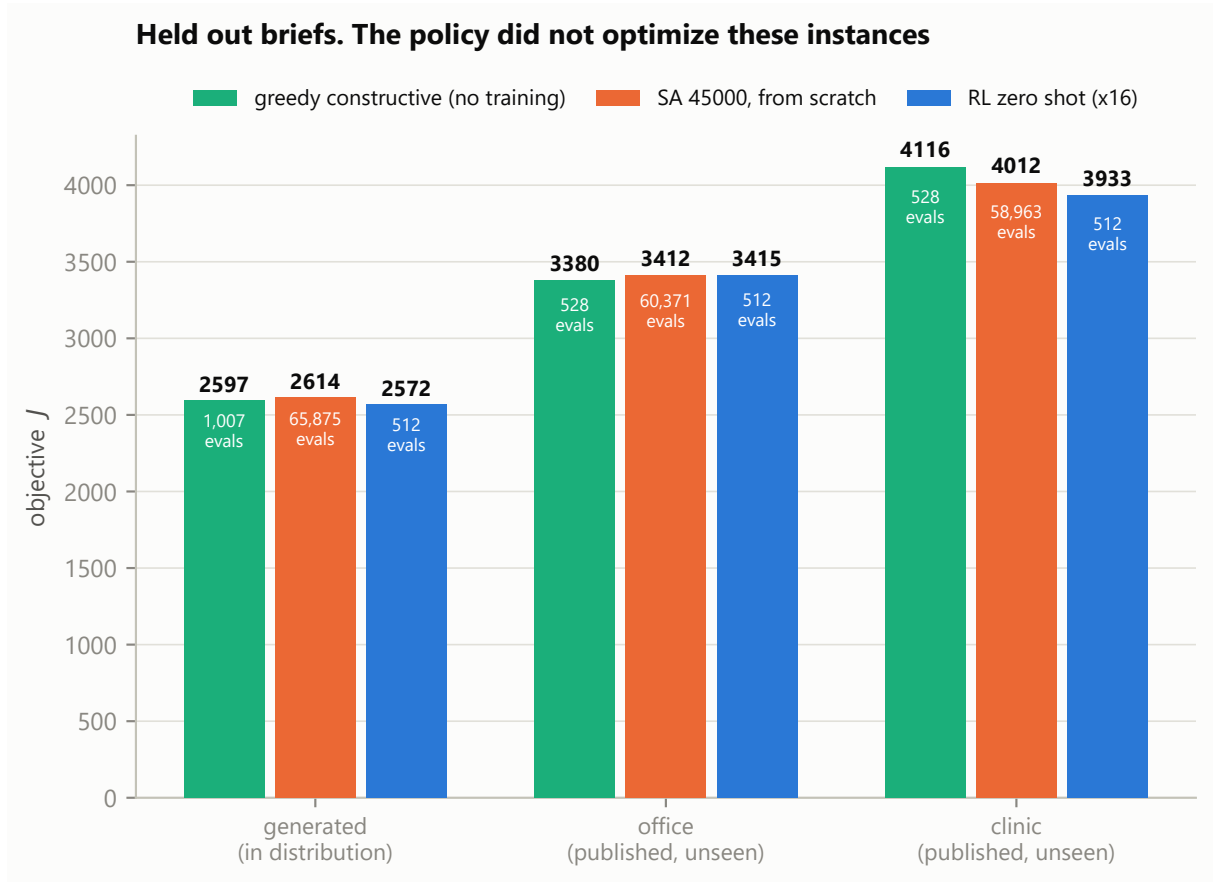


Figure S1: Held out briefs, annotated with objective evaluations per layout. The two reference scenarios were authored by hand and never optimized in training. The RL bars show 16 samples without the final polish (512 evaluations), a cheaper operating point than the $\times 16$ +polish column of Table S1, which is why the two differ slightly.

3 955.6 against the policy’s 3 848.5. Across the three test sets the zero shot policy is better on one, similar on one and worse on one. These results show that one set of weights can evaluate unseen briefs with a quality similar to large budget annealing while using three to four orders of magnitude less search. They do not show that the policy is always better. Transfer is evaluated at a fixed facility count (main text, Limitations).

S2 Quality and diversity

A common argument for learning in design is that a policy provides a *distribution* over layouts. Previous comparisons used one policy temperature and several annealing budgets. We vary all three methods. We vary the sampling temperature for the policy, the budget for annealing, and the fraction of randomized commitments for the constructor without training.

Table S2: Quality against diversity on OFFICE, 64 samples from one start. Diversity is the mean pairwise fraction of cells whose room type differs, estimated from 400 of the 2016 pairs, so it carries about ± 0.01 of sampling noise; every row uses the same estimator.

method	J	diversity	ms/sample
policy, order chosen by policy, $T=1.0$	3 529.3	0.006	22
policy, random commit order, $T=1.0$	3 508.0	0.304	19
$T=2.5$	3 418.2	0.405	18
$T=5.0$	3 048.0	0.623	13
SA 8 192, lattice	3 321.1	0.581	478
SA 45 000, lattice	3 414.5	0.639	1 070
greedy constructive, area order	3 359.5	0.000	7
20% of commit slots reshuffled	3 374.5	0.350	114
50%	3 408.3	0.482	251
70%	3 415.2	0.531	294
100%	3 405.2	0.620	347

We make three observations. First, a policy whose entropy bonus is annealed to zero produces almost no diversity. At $T=1$, its diversity is 0.006. Second, randomizing the commit order increases diversity by a factor of fifty with a 0.6% decrease in objective. Third, applying the same procedure to the greedy rule produces its own front. Randomizing a controlled fraction of the commit slots changes diversity continuously from 0.000 to 0.620 and reaches 3 415.2 at diversity 0.531 for 32 objective evaluations, which matches the best annealing point anywhere on the front (3 414.5 at 0.639, 61 267 evaluations).

The two fronts cross once. The policy has a higher objective up to diversity ≈ 0.42 , while the constructor without training has a higher objective above ≈ 0.45 . An earlier, coarser sweep of the heuristic, which tried only a uniform random order and therefore jumps from 0.000 to 0.603, suggested a gap in its range that does not exist. This is a single scenario result and we did not repeat it on the suite.

S3 Amortized cost

Training has a fixed initial cost, while search has a cost for every layout. Table S3 reports the annealing budget sweep on OFFICE, each budget’s wall clock time per layout, and the first point

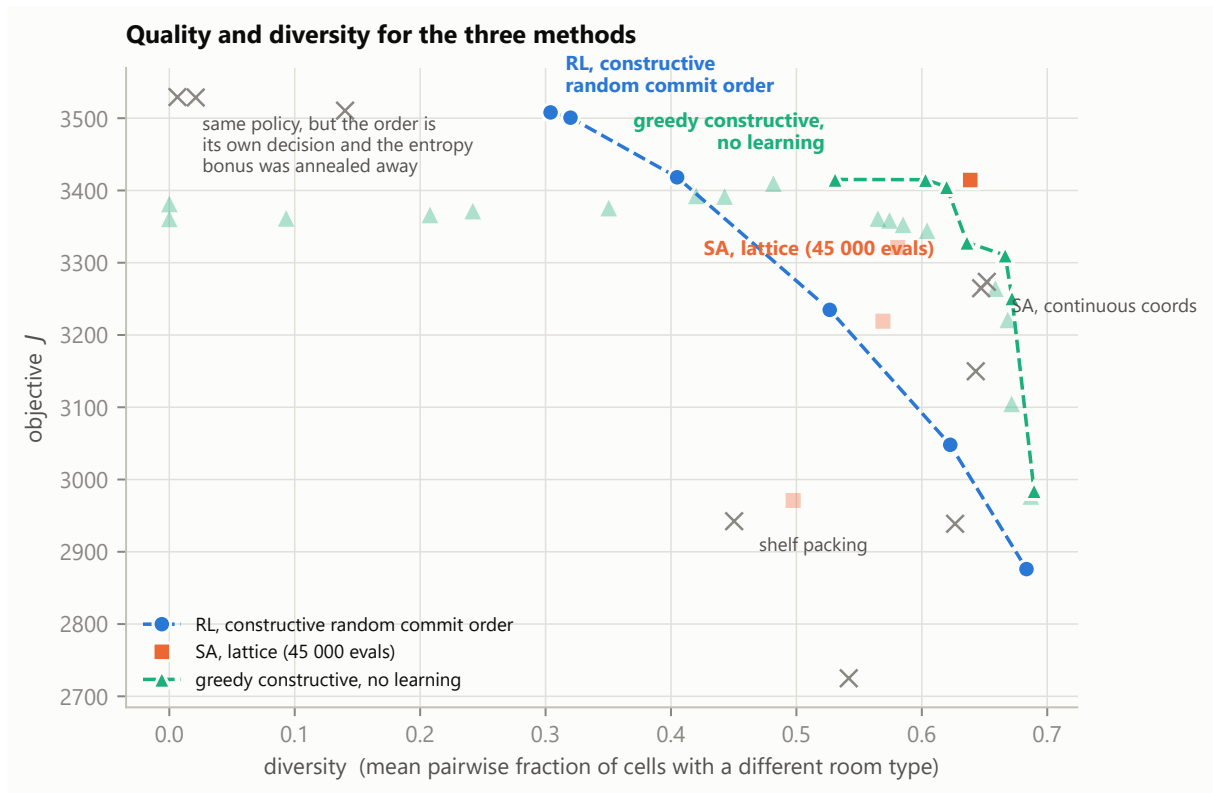


Figure S2: Quality against diversity with all three sides swept. Faint markers are points that a method's own front dominates.

on the policy’s training curve (488 recorded checkpoints, 2049s total) at which the policy’s greedy decode exceeds that budget’s mean objective. The policy finishes at 3 529.3 and decodes in 17.8ms per layout.

Table S3: Annealing budget against the training time needed to match it. The 45k lattice row is the measurement of the main comparison (the budget sweep’s own 45k run gives 3 400.8 at 1.13s); the larger budgets come from the dedicated sweep. The legacy scenario implementation is not batched, so its time per layout is omitted.

annealer	mean J	s/layout	training s to exceed
legacy scenario implementation, 45k	3 269.9	not reported	109
lattice, 45k	3 414.8	1.00	157
lattice, 150k	3 459.0	3.40	215
lattice, 400k	3 489.1	8.83	256
lattice, 10^6	3 501.3	20.08	268

Charging the full 2049s up front and 17.8ms per layout thereafter, the policy has a lower total wall clock cost than annealing with 10^6 evaluations after $2049/(20.08-0.018) \approx 102$ layouts. Compared with lattice annealing using 45 000 evaluations, this occurs after $2049/(1.00-0.018) \approx 2 093$ layouts.

This comparison has two limitations. First, the workload contains repeated requests for the *same* brief. An annealer could store its best layout and reuse it for later requests. A stream of unseen briefs is evaluated separately in S1. Second, this comparison is against annealing and not against the constructor without training. At 88ms per layout, the initial training cost is recovered only after roughly 29 000 layouts relative to the constructor. Thus, amortization does not provide an advantage over this constructor for smaller workloads.

S4 Action granularity

We compare a discrete action head for small moves with a continuous displacement head when feasibility is handled by a penalty. Our earlier experiments on a small scenario suggested a tradeoff between feasibility and return, but they were conducted before the present protocol and are not used as evidence. We report the comparison inside the corrected improvement MDP on the single $n = 32$ OFFICE scenario (best observed reward, jump and swap moves, Metropolis acceptance). One checkpoint per action head is evaluated from the same 64 independent random starts, shown in Table S4. No tradeoff appears in this evaluation. With sampled decoding at horizon 4 096 the lattice head reaches 3 069.6 at 50% feasibility while the continuous head reaches 2 737.0 at 0%, and deterministic decoding of either head falls into a limit cycle and loses more than a thousand objective units. The feasible-volume record in Table S12 explains the direction geometrically, and under the constructive formulation the question does not arise.

S5 Improvement formulation ablation, in full

The main text summarizes the corrections to the improvement MDP on OFFICE; Table S5 gives every variant. Each is trained for a matched budget of 688k environment steps and evaluated on 64 held out starts using stochastic decoding at 4 096 objective evaluations.

Table S4: Discrete against continuous action head inside the corrected improvement MDP on OFFICE, $n = 32$: mean and standard deviation across the same 64 independent evaluation starts.

head	decode	horizon	J	feasible
discrete (lattice)	sampled	512	$2\,475.3 \pm 93.8$	0.00
discrete (lattice)	sampled	2\,048	$2\,853.0 \pm 101.4$	0.06
discrete (lattice)	sampled	4\,096	$3\,069.6 \pm 126.3$	0.50
discrete (lattice)	argmax	4\,096	$1\,372.6 \pm 228.4$	0.00
continuous	sampled	512	$2\,354.7 \pm 76.2$	0.00
continuous	sampled	2\,048	$2\,611.4 \pm 85.5$	0.00
continuous	sampled	4\,096	$2\,737.0 \pm 60.8$	0.00
continuous	mean	4\,096	$1\,649.5 \pm 203.9$	0.00

Table S5: Improvement formulation ablation on OFFICE.

ID	configuration	J	feasible
A0	reference: difference reward, translate, $\gamma=0.995$	2\,658.5	0 %
A0'	the same with $\gamma=0$	1\,804.3	0 %
A1	+ best observed reward	2\,246.1	0 %
A2	+ best observed reward, jump and swap	2\,461.8	0 %
C1	+ Metropolis accept/reject (lattice head)	3\,067.2	48 %
C1'	the same with a continuous head	2\,737.0	0 %
A0 evaluated deterministically (argmax)		1\,445.5	0 %

S6 Certificate gallery

Figure S3 shows one held out instance at fill ratio 0.95 solved by the same largest first constructor under the four certificate settings of the main text’s certificate comparison. Without a certificate the constructor occupies too much free space in the early steps, and the last facilities have no legal position; the scalar objective reports this only as a lower number.

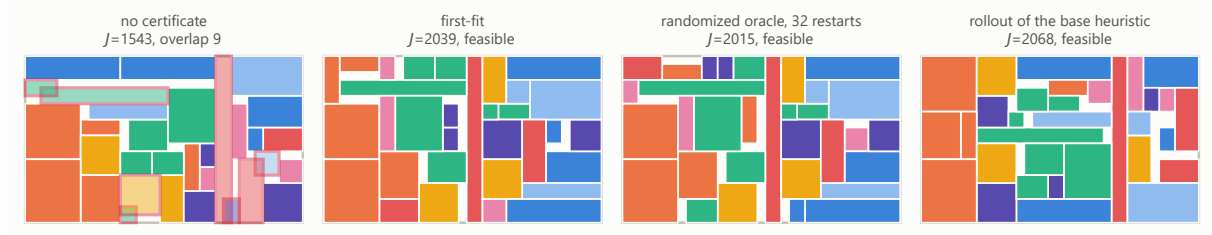


Figure S3: One instance, four certificates, same constructor. Facilities that end up overlapping are outlined in red.

S7 MCNC/GSRC conversion, in full

HPWL numbers depend on the conversion, so we state it completely. Block dimensions come from the GSRC `.blocks` files (`hardrectilinear` records, width and height from the corner list); netlists from the UCLA `.nets` files; terminal positions from the `.pl` files, matched to net pins by name. Pins are placed at block centers and terminals at their given coordinates, the standard

simplification. Nets with fewer than two pins are dropped, contributing zero. Coordinates are divided by an integer for each instance and floored, which is necessary because the legality masks are dense over the plate: the factor is 6 for AMI33, 33 for AMI49 and 3 for the GSRC instances, chosen as the smallest divisor bringing the plate under 200 cells per side, and it is reported with every result. The outline is fixed by area, $\text{plate} = \text{total block area} / (1 - \text{dead space})$, at aspect ratio 1; blocks may be rotated by 90° . Under this conversion HPWL is comparable across the paper’s own methods but not against published floorplanning numbers, which use full resolution and a different pin model. Integer flooring changes block shapes and packing possibilities, so converted dead space is not treated as a published full-resolution result.

S8 Larger instances

Table S6 gives the full feasibility record of the constructors without training on 96 to 256 facilities summarized in the main text.

Table S6: Constructors without training on larger instances, feasibility rate. Every instance has a verified feasible packing, so a failure indicates that the method did not find the existing packing. Plate geometry is drawn per cell, so rows are not comparable with one another; the comparison is between methods within a row. The rollout certificate is $O(n^2WH)$ per query and was not run above 128 facilities; that is recorded rather than omitted.

n	instances	no certificate	first fit	rollout
<i>fill ratio 0.85</i>				
96	15	1.00	1.00	1.00
128	15	1.00	1.00	1.00
192	15	1.00	1.00	not run
256	15	1.00	1.00	not run
<i>fill ratio 0.95</i>				
96	30	0.57	0.53	0.90
128	30	0.53	0.57	0.90
192	30	0.70	0.63	not run
256	30	0.83	0.83	not run

S9 Complete results for methods starting from the same layout

Every row below uses a common verified best-contact layout and retains it as a feasible incumbent. Stochastic seeds are averaged within an instance before cell means are computed. The 0.1 and 0.3 second values are read from the same anytime run that continues to 1.0 second. Times exclude the common measured initialization cost, which is charged equally to all methods in total-time reporting. Mean initialization and verification time ranges from 0.0052 to 0.0128 seconds over the four cells. All normalized improvements use the full feasible objective J_β , including the constant no-overlap bonus.

At 1.0 second, the raw M1 minus B1 difference is 224.5 objective units [172.2,276.5], its win/tie/loss count is 156/0/40, and 152 of 196 instances exceed the one-percent practical threshold. The corresponding counts against B2, B2S, M0 and R0 are 196/0/0, 195/0/1, 109/0/87 and 195/0/1. The M0 interval does not establish a difference.

Table S7: Dense confirmation at fill 0.95. Entries are mean percentage improvement over the common initial layout.

geometry	n	sec.	B1	B2	B2S	B3	M0	R0	M1
guillotine	32	0.1	1.72	0.55	0.82	2.45	3.50	1.64	6.14
		0.3	1.72	1.22	1.82	4.37	7.59	1.96	7.83
		1.0	1.72	2.40	3.51	9.63	9.04	2.43	9.51
guillotine	64	0.1	0.00	0.22	0.41	1.29	1.72	0.21	1.41
		0.3	4.20	0.68	0.87	2.21	3.64	1.33	5.21
		1.0	4.44	1.41	1.62	3.69	8.00	2.31	8.76
nonslicing	32	0.1	1.79	0.57	1.02	1.95	3.50	1.91	6.32
		0.3	1.79	1.43	2.03	4.13	7.63	2.27	7.83
		1.0	1.79	2.64	3.71	9.53	10.15	2.86	9.65
nonslicing	64	0.1	0.00	0.18	0.44	1.64	1.47	0.13	1.14
		0.3	3.46	0.74	0.96	2.31	2.56	1.46	4.68
		1.0	4.10	1.38	1.78	3.60	7.78	2.70	7.98

Table S8: Pooled M1 minus control contrasts on 196 initialized dense instances. Entries are percentage points with paired instance-bootstrap 95% intervals.

sec.	B1	B2	B2S	B3	M0	R0
0.1	2.84 [2.26,3.45]	3.32 [2.83,3.82]	3.03 [2.56,3.52]	1.87 [1.36,2.39]	1.18 [0.58,1.75]	2.75 [2.33,3.17]
0.3	3.55 [2.57,4.49]	5.35 [4.84,5.86]	4.95 [4.47,5.43]	3.12 [2.58,3.67]	1.05 [0.46,1.67]	4.61 [4.12,5.10]
1.0	5.93 [4.99,6.88]	7.02 [6.54,7.52]	6.33 [5.84,6.82]	2.41 [1.74,3.06]	0.25 [-0.42,0.92]	6.39 [5.89,6.92]

Table S9: Boundary study at fill 0.90 and 1.0 second. Entries are mean percentage improvement over the common initial layout. The last column is the paired M1 minus B1 percentage-point difference with a 95% bootstrap interval.

geometry	n	init.	B1	B2	B2S	B3	M0	R0	M1	M1-B1
guillotine	32	20/20	13.03	7.89	7.60	8.84	16.17	7.97	14.60	1.57 [-0.27,3.65]
guillotine	64	20/20	14.06	5.53	4.94	4.70	15.48	6.25	13.44	-0.62 [-2.09,1.07]
nonslicing	32	19/20	11.86	6.92	7.28	9.43	14.90	8.03	13.84	1.97 [-0.07,4.02]
nonslicing	64	20/20	14.98	5.01	4.82	4.10	15.85	6.68	13.07	-1.91 [-3.54,-0.25]

Table S10: Mechanism log at fill 0.95 and 1.0 second. Repair success is conditional on a nonempty displaced set with size at most four. Verified completions exclude the carried fallback. R0 contains reuse only, and M0 uses full regeneration for every nonfallback proposal.

geometry	n	mean $ D $	repair success	M1 reuse	M1 verified	R0 verified	M0 verified
guillotine	32	1.95	2.45%	14.0	80.8	26.4	59.6
guillotine	64	2.49	3.91%	6.4	61.1	23.8	40.9
nonslicing	32	2.05	2.19%	15.9	76.1	29.7	59.0
nonslicing	64	2.72	3.55%	6.9	55.0	26.2	35.7

Across all dense M1 runs, the repair branch produces 6,215 incumbent updates and 142,270 objective units of logged gain; unchanged reuse produces 1,380 updates and 12,455 units. These event counts are mechanism descriptions rather than inferential samples. B2 and B2S have mean destroy sizes 8.88 to 9.06 and 2.99, respectively.

Table S11: Mean measured work per one-second dense run, in milliseconds. Late is the number of candidates that completed after the cutoff and were discarded.

method	generation	intermediate verify	final verify	objective	return time	late
B1	0.0	0.0	0.1	0.1	1001.53	0
B2	0.0	0.0	2.9	2.6	1002.58	1
B2S	0.0	0.0	15.6	13.5	1002.67	14
B3	845.4	37.7	40.3	40.9	1001.44	28
M0	719.3	21.4	30.7	30.6	1001.26	23
R0	0.0	12.6	105.0	92.9	1001.66	81
M1	358.1	30.3	65.2	61.8	1001.11	47

All 8,820 dense budget layouts and 5,214 boundary budget layouts pass an independent result audit. Recomputed objective and trace selection errors are zero. No admitted dense event is later than 0.999998 seconds. Nonpreemptive work can return up to 9.8 ms after the cutoff, but its candidate is not admitted. The adversarial regression set rejects duplicate and missing facilities, negative and out-of-range coordinates, overlap, stale witnesses and incomplete timeout returns, while accepting legal edge contact.

S10 Diagnostic evidence omitted from the focused main text

These experiments motivated completion-aware construction but are not evidence for the primary matched-time claim. Table S12 shows why an overlap penalty receives little feasible exploration as size and density increase. The lattice column is higher than the continuous-center column wherever both are resolved, so action representation matters before learning.

Table S12: Fraction of uniformly sampled layouts that are overlap free. Each entry averages four generated instances at 4×10^4 samples; $< 6 \times 10^{-6}$ means no feasible sample was drawn.

n	fill 0.35		fill 0.50		fill 0.65	
	continuous	lattice	continuous	lattice	continuous	lattice
2	2.6×10^{-1}	3.7×10^{-1}	1.7×10^{-1}	2.4×10^{-1}	7.1×10^{-2}	1.4×10^{-1}
4	3.3×10^{-2}	7.6×10^{-2}	3.0×10^{-3}	1.5×10^{-2}	5.0×10^{-4}	3.9×10^{-3}
6	4.0×10^{-3}	1.7×10^{-2}	1.6×10^{-4}	1.1×10^{-3}	$< 6 \times 10^{-6}$	2.5×10^{-5}
8	5.9×10^{-4}	4.5×10^{-3}	1.9×10^{-5}	6.9×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
10	8.8×10^{-5}	1.1×10^{-3}	$< 6 \times 10^{-6}$	1.3×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
12	6.3×10^{-6}	2.2×10^{-4}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$
16	$< 6 \times 10^{-6}$	1.3×10^{-5}	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$	$< 6 \times 10^{-6}$

The earlier single-facility improvement MDP used the reward $J(s_{t+1}) - J(s_t)$. With discount one it optimizes the terminal objective; with discount below one it favors high intermediate values and penalizes recovered detours. At a common 688k-step training budget, adding a best-observed reward, jump and swap moves, and Metropolis acceptance raised the objective from 2658.5 to 3067.2 and feasibility from 0% to 48%. Lattice annealing reached 3414.8, and a

masked constructor without training reached 3380.2. Table S5 gives the complete formulation ablation.

Initialization was also a major confound in the old decoder. On 100-instance dense cells $(n, f) = (32, 0.90), (32, 0.95), (64, 0.95)$, bottom-left fill in area order covered 0.78, 0.09 and 0.28; best-contact covered 0.98, 0.96 and 1.00. This is why every primary method now receives the same archived best-contact layout. The legacy exact-marginal selector matched or exceeded the two learned selectors in most $n = 64$ cells, so learning is not used in M1.

Finally, the learned completion predictor was trained on 25,258 rollout states with an instance-grouped split. Validation AUC was 0.902, but on disjoint fill 0.95 tests its best completion rate was 0.20, compared with 0.90 for verified first fit and 1.00 for the more expensive rollout certificate. The labels record success of one heuristic rollout, not true membership in the set of completable states. This negative result supports learned candidate ordering, but not using an unverified classifier as a hard feasibility decision.